

Simulation and Monte Carlo

2025-2026 projects

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You must form a group of **three** students **among** your TD class, select one of the following projects, and do it by the last TD class, where you will do an oral presentation of your results in front of the other students. You may return a document with your plots and results to your “chargé(e) de TD” (no need to return a formal report). Please make sure you also send your code.

Important: Always make sure you have assessed in some way the numerical error of your estimates (e.g. through box-plots over repeated runs for instance).

Bonus questions are **optional**: do them when are confident you treated properly the previous points.

It’s ok to look up online for solutions (and adapt code you found on the internet), but cite your sources accordingly.

If you are stuck, feel free to get in touch with your “chargé(e) de TD”, who will let me know.

Poisson GLM

Consider a Poisson GLM (generalized linear model), where, given a vector $x_i \in \mathbb{R}^p$ of predictors, the outcome variable Y_i follows a Poisson distribution, with parameter $\lambda_i = \exp(\beta^T x_i)$, $\beta \in \mathbb{R}^p$. The first component of each x_i is conventionally 1, i.e., it is an intercept. The Y_i are independent, $i = 1, \dots, n$. We assign a Gaussian $N(0, \tau I)$ prior to β , with $\tau = 10^3$. The x_i are pre-processed to have mean zero and variance one.

1. Derive the likelihood of the model. Explain how one may derive a Gaussian approximation of the posterior using optimisation (e.g., Newton), as seen in the course for the probit model.

2. Simulate data from the model. Implement the Gaussian approximation of the previous question, and use it in some way to implement (a) importance sampling, and (b) random walk Metropolis to estimate expectations with respect to the posterior distribution. Discuss how p and n affect the relative performance of both methods.
3. Apply both algorithms to a real dataset such as the **French Motor Third-Party Liability Claims** (available in scikit-learn), and discuss.

Nested sampling

The aim of this project is to implement the nested sampling (NS) algorithm. You may read the following paper for a succinct description of NS: <https://arxiv.org/abs/0801.3887>.

We consider a Bayesian model where $y_i \sim N_d(\theta, \Sigma)$, $i = 1, \dots, n$, Σ is fixed (and known), and the vector $\theta \in \mathbb{R}^d$ has prior $N_d(0_d, I_d)$. The prior density is denoted by $\pi(\theta)$, and the likelihood by $L(\theta)$ (which depends implicitly on y , $y = (y_1, \dots, y_n)$). You can sample y from the model.

1. Derive the posterior distribution of θ , and the marginal likelihood $p(y)$.
2. Assume first that $\bar{y} = (y_1 + \dots + y_n)/n = 0_d$ and $\Sigma = I_d$. Show that it is possible to sample exactly θ from the prior, conditional on $L(\theta) > l$, for an arbitrary $l > 0$. Implement NS in this case, and assess its performance (i.e. how well does it approximate the marginal likelihood).
3. We consider now the general case. Show that it is possible to implement an MCMC sampler to sample from the same type of distribution (prior constrained to $L(\theta) > l$.) Explain how you can use this MCMC sampler within NS. Implement the corresponding algorithm.
4. Assume now that Σ is also unknown and must be estimated. Explain how to change the prior distribution so that the posterior remains tractable. (Search the literature for conjugate priors for a general multivariate Gaussian model.) Adapt your NS algorithm to this setting. Note: it's ok if, for this question, the method works reasonably well only for small d .

Likelihood-free inference and sums of log-normal variates

This project is partly inspired by the following article: <https://arxiv.org/abs/1701.05146>, by Bernton et al, 2017, in particular Section 4.2. The goal is to estimate the parameters $\theta = (\mu, \sigma^2)$ based on observed data: $Y_i = \sum_{l=1}^L \exp\{X_{i,l}\}$, $X_{i,l} \sim N(\mu, \sigma^2)$ where the $X_{i,l}$ are IID (independent and identically distributed). In words, Y_i is the sum of $L \geq 2$ log-normal variates; there is no closed-form expression for the density of the Y_i 's.

1. Explain how to use Reject-ABC to estimate θ , and how the choice of the prior, and ϵ affect the result (where ϵ is defined as in the course). Note: You should generate first ‘true’ data from the model itself, using e.g. $L = 10$, $\mu = 0$, $\sigma = 0.3$. You may use the same distance (MEWE) between the simulated and real data as in the paper. For the prior, you may take for instance $\mu \sim N(0, s^2)$, $\log \sigma^2 \sim N(0, t^2)$, but then discuss the impact of s and t on the performance of the algorithm.
2. Show it is possible to derive a MCMC sampler that targets the same distribution as in point 1, and that this MCMC sampler may perform better (in some sense that you will discuss) for certain values of ϵ and/or certain choices of the prior. Hint: look for “MCMC-ABC” (Marjoram et al) on the web.
3. It is actually possible to derive an exact MCMC sampler that simulates from the posterior distribution of θ for this model. The idea is to treat the $X_{i,l}$ as latent variables, in the spirit of “data augmentation”, as discussed in the course. Explain how you can rewrite the joint target distribution for $(\theta, (X_{i,l}))$ to that effect, and derive a Gibbs sampler that (a) samples from θ given the $X_{i,l}$ (this conditional distribution has a close form); and (b) samples blocks of two variables $(X_{i,l}, X_{i,l'})$ using a Metropolis step. The pair (l, l') may be chosen randomly. It may make sense to perform (b) for a certain number of pairs, before doing step (a) again.
4. Use the Gibbs sampler of Question 3 to assess the bias introduced by ABC for different values of ϵ .

Gibbs and ABC

This project is based on the following recent paper: <https://arxiv.org/abs/1905.13599> and in particular the moving average (toy) example in the supplement.

Your goal is to compare three algorithms:

1. The Gibbs-ABC algorithm proposed in the paper;
2. A more standard ABC sampler. You can simply consider ABC-reject, but it could be interesting to consider a more sophisticated sampler;
3. A random walk Metropolis sampler (note that this is a toy example where the likelihood is actually tractable, so we can use such a standard, non-ABC approach here).

Think carefully about how to make this comparison scientifically relevant and interesting. In particular, note that 3 should be a “gold standard” (why?), and that Algorithms 1 and 2 should be compared in terms of three metrics: CPU time, inferential error, and Monte Carlo error.

As a bonus, you can consider another model (from the paper or from another paper on ABC).

Gibbs measures and Monte Carlo integration

In Statistical Physics, Gibbs measures are of the form

$$\pi(x_1, \dots, x_n) \propto \exp \{-\beta H_n(x_1, \dots, x_n)\}$$

where the energy H_n is a function of n points $x_i \in \mathcal{X}$, depends on a potential function $V : \mathcal{X} \rightarrow \mathbb{R}$ and an interaction kernel $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}^+$, as follows:

$$H_n(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n V(x_i) + \frac{1}{2n^2} \sum_{i \neq j} K(x_i, x_j).$$

1. Propose a MCMC algorithm to sample from π , and implement it. Explain properly how you need to calibrate the algorithm to optimise the mixing of the chain, and how the performance depends on β . You may take $\mathcal{X} = \{x \in \mathbb{R}^2, \|x\| \leq 1\}$, $V(x) = x^2/2$, and $K(x, y) = \exp\{-c\|x - y\|\}$ for different values of c .
2. Martin Rouault (an ENSAE alumnus) proposed in his PhD to use Gibbs measure to generate samples of size n such that (a) $n^{-1} \sum_{i=1}^n \varphi(x_i)$ converges to $\pi(\varphi)$ (the expectation of $\varphi(X)$ for $X \sim \pi$), (b) the x_i ‘repel’ each other, to better cover the space (a bit in the spirit of low-discrepancy sequences). For this to happen, one needs to take $V(y) = \int \pi(x) K(x, y) dy$, and, say $K(x, y) = -\log \|x - y\|$, assuming $\mathcal{X} \subset \mathbb{R}^d$, $d = 1$ or $d = 2$. This poses a practical problem: how can we evaluate V , given that it’s already an expectation with respect to the target distribution π ? MR suggested to replace V by an empirical estimate, based on preliminary sample of IID variables simulated from π . Investigate this approach. You may take $d = 2$, π a bivariate Gaussian $X \sim N(0, I_2)$, truncated to $\|X\| \leq 1$. Explain how you can sample independently from this distribution.
3. Replicate the Bayesian classification experiment of the following paper: <https://arxiv.org/pdf/2508.01392> (Keep in mind that the proposed approach is supposed to work only for distributions that are defined on a compact set. In practice, you may replace the target distribution by a truncated version. Also, note that for $d \geq 3$, the kernel has a different expression, as explained in the paper.)

Control variates for a GARCH model

This project is based on the numerical example on GARCH modelling from this paper: <https://doi.org/10.1007/s11222-012-9344-6>.

1. Follow the guidelines given in that experiment to implement a random walk Metropolis sampler that targets the posterior distribution of a GARCH model. For the data, you can use simulated data at first, and then look at the same type of real data (log-returns computed from exchange rates) in a second time.
2. Consider the control variates proposed in the paper (Equation (8), and in particular the versions based on a first-order polynomial, i.e. $f(x) + a^T z$ on the same page as (8)). Explain how you can use these control variates in this settings by doing a linear regression.
3. Consider a larger set of control variates (by using a polynomial of order 2 for instance). When the number of control variates become too large, the linear regression approach of Step 2 may become too expensive (explain why). Adapt the method proposed in this paper: <https://doi.org/10.1007/s11222-021-10011-z>, which relies on the Lasso, to your problem, and compare with the naive approach (where you use only a single step based on a linear regression).
4. Bonus: is the fact that we do linear regression completely valid when applied to a MCMC sample? Try to think of ways to address this issue.

Hint: you could think of ways to make MCMC simulations less “dependent”, by sub-sampling, or averaging over blocks of a given size, or another approach.